1 Rainy Day Scenario Analysis

Table 1: Rainy Day Scenario Comparison (Flash & Pro)

		Fuzhou			Guangzhou			Shenzhen			Shanghai		
Model	Metric	ROAM	Bare	Baseline	ROAM	Bare	Baseline	ROAM	Bare	Baseline	ROAM	Bare	Baseline
Flash	POI↑	33	18	16	33	18	19	24	15	15	15	20	18
	SM↑	0.3	0.3	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	TT↓	5.6	3.4	0.0	7.0	4.4	0.0	6.4	5.6	0.0	2.4	3.8	0.0
	CE↑	**0.4**	**0.4**	0.0	**1.0**	0.8	0.6	**5.3**	1.7	0.6	0.3	**0.7**	0.4
	DB	0.0	0.2	0.1	0.0	0.2	0.1	0.0	0.0	0.0	0.6	0.1	0.0
	RC	454.4	421.9	1360.4	555.5	536.0	1905.1	620.3	853.2	1586.5	316.6	428.3	1371.4
Pro	POI↑	17	19	17	16	17	17	33	16	16	15	18	18
	SM↑	0.3	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	TT↓	3.2	4.4	0.0	4.7	4.9	0.0	9.6	5.0	0.0	7.3	3.6	0.0
	CE↑	0.1	**0.4**	0.1	0.6	**3.7**	0.5	**32.2**	9.7	0.5	**3.3**	0.6	0.7
	DB	0.1	0.1	0.2	0.1	0.1	0.2	0.0	0.1	0.1	0.0	0.3	0.2
	RC	402.4	507.0	686.3	643.6	635.1	1776.2	1072.5	720.3	1120.4	1009.9	460.5	1915.7

2 Full Detailed Results

Table 2: Full Detailed Results (Flash)

		Fuzhou			Guangzhou			Shenzhen			Shanghai		
Query	Metric	ROAM	Bare	Baseline	ROAM	Bare	Baseline	ROAM	Bare	Baseline	ROAM	Bare	Baseline
Sunny	POI↑	28	21	16	21	20	19	28	17	18	33	19	15
	SM↑	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.3	0.4	0.3	0.4	0.4
	TT↓	7.6	3.9	0.0	**6.8**	8.2	0.0	12.2	6.9	0.0	6.6	3.2	0.0
	CE↑	0.5	0.3	1.0	0.7	4.8	0.6	27.7	37.5	1.2	1.1	0.4	0.2
	DB	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.0	0.0	0.1	0.0
	RC	554.8	411.8	3856.9	626.3	869.6	1433.9	1043.2	872.4	1716.1	483.3	379.0	1396.4
Rainy	POI↑	33	18	16	33	18	19	24	15	15	15	20	18
	SM↑	0.3	0.3	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	TT↓	5.6	3.4	0.0	7.0	4.4	0.0	6.4	5.6	0.0	2.4	3.8	0.0
	CE↑	**0.4**	**0.4**	0.0	**1.0**	0.8	0.6	**5.3**	1.7	0.6	0.3	0.7	0.4
	DB	0.0	0.2	0.1	0.0	0.2	0.1	0.0	0.0	0.0	0.6	0.1	0.0
	RC	454.4	421.9	1360.4	555.5	536.0	1905.1	620.3	853.2	1586.5	316.6	428.3	1371.4
Sightseeing	**POI↑**	**36**	16	18	**35**	15	19	**34**	15	16	**33**	16	16
	SM↑	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5
	TT↓	8.7	6.7	0.0	12.0	5.2	0.0	8.6	4.5	0.0	13.8	3.8	0.0
	CE↑	1.6	0.7	0.0	6.1	2.3	1.1	9.8	4.4	0.5	29.7	0.4	0.1
	DB	0.0	0.1	0.2	0.0	0.0	0.2	0.1	0.0	0.1	0.2	0.1	0.1
	RC	710.8	933.2	526.3	889.1	786.9	1932.6	720.2	675.0	1765.6	1653.1	550.6	1528.8
City Exploration	POI↑	24	20	19	23	18	15	25	15	18	21	19	16
	SM↑	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.3	0.3
	TT↓	4.8	3.6	0.0	4.9	4.1	0.0	7.6	4.6	0.0	5.7	3.4	0.0
	CE↑	**0.3**	**0.3**	0.1	**0.7**	0.6	0.3	5.4	16.4	2.9	**0.7**	0.6	0.2
	DB	0.7	0.1	0.2	0.1	0.0	0.0	0.1	0.2	0.2	0.0	0.1	0.1
	RC	404.0	392.6	621.1	426.9	505.9	1436.6	583.5	704.6	2095.4	514.6	403.5	1415.6

3 Evaluation Metrics

POI —Point of Interest (# of POIs, ↑ higher is better) $\text{POI} = \sum_{\text{days}} |\text{itinerary}_d|$. Total number of POIs arranged in the itinerary, reflecting route richness and information density.

SM —Semantic Matching (cosine similarity, ↑ higher is better) $\text{SM} = \frac{1}{N}\sum_{i=1}^{N} \cos(\mathbf{e}_{\text{poi}_i}, \mathbf{e}_{\text{query}})$. Average cosine similarity between selected POI text embeddings and the user query embedding, evaluating semantic consistency between itinerary content and user preferences.

TT —Transit Time Ratio (%, ↓ lower is better) $\text{TT} = \frac{\sum \text{commute time (min)}}{\text{available minutes} \times \text{days}} \times 100\%$. Percentage of commute time relative to total available daily minutes, reflecting time utilization efficiency. Lower TT indicates more compact connections between POIs.

CE —City Exploration Rate (%, ↑ higher is better) $\text{CE} = \frac{\text{ConvexHull(trip POI coordinates)}}{\text{ConvexHull(city-wide POI coordinates)}} \times 100\%$. Ratio of the convex hull area spanned by the trip's POIs to that of all city POIs, measuring the spatial exploration breadth of the route. Higher values indicate broader city coverage.

DB —Day Balance (coefficient of variation, lower is more balanced) $\text{DB} = \text{CV} = \frac{\sigma}{\mu}$, where σ is the standard deviation and μ is the mean of daily POI counts. Coefficient of variation of POI counts across days. Lower DB indicates a more evenly distributed daily load, avoiding over-crowded or under-filled days.

RC —Route Compactness (weighted meters per segment, lower is more compact) $\text{RC} = \frac{\sum \text{weighted distance}_s}{\text{total POIs} - \text{days}}$, where weighted distance $= \text{walk}_m \times 1.0 + \text{bike}_m \times 0.32 + \text{car}_m \times 0.16$. Average walking-equivalent distance per segment between adjacent POIs. Uses time-normalized weights: driving covers ground quickly and should not be misjudged as sparse, while walking distance faithfully reflects true compactness. Lower RC means less travel cost between POIs.